Football Match Outcome Prediction

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January 2026

Introduction

Predicting the outcomes of football matches is a complex challenge due to the high uncertainty of the game and the influence of numerous contextual factors. In this project, a system is developed to anticipate the final result of matches—home win, draw, or away win—using Machine Learning and Deep Learning techniques, supported by an extensive historical dataset. The proposed model combines advanced team statistics, contextual competition information, and sentiment analysis extracted from post-match reports, incorporating subjective variables to enhance predictive performance compared to traditional approaches.

For training and evaluation, data from the five major European leagues over the last nine seasons—La Liga, Premier League, Serie A, Ligue 1, and Bundesliga—has been used, comprising approximately 17,000 matches. The dataset integrates performance statistics at both team and player levels for each match. As a result, the developed model achieves an accuracy of 57 %, with the main bottleneck being the prediction of the draw class. These results outperform baseline approaches and demonstrate the viability of combining quantitative and qualitative information in sports outcome prediction.

Data and Methodology

Data

The data used in this project has been collected from publicly available sources specialized in football analytics. Specifically, match statistics were obtained from *Sofascore*, while detailed squad information was sourced from *Transfermarkt*. The resulting dataset integrates performance information at both team and player levels for each participating team.

To incorporate qualitative information, match reports were collected from *WhoScored* and processed into sentiment-based features.

As mentioned earlier, the dataset spans the five major European leagues from the 2015–2016 season to the 2023–2024 season, resulting in approximately 17,000 matches and 250 variables.

Methodology

The methodology is based on comparing different modeling approaches for predicting match outcomes. Three deep learning architectures were developed and evaluated: a Dense Neural Network (DNN), a Long Short-Term Memory network (LSTM), and a Transformer-based model.

For the DNN model, preprocessing tailored to tabular data was applied, including missing value handling, normalization of numerical variables, dimensionality reduction using Principal Component Analysis (PCA), and feature aggregation based on the previous five matches. This approach condenses relevant statistical information into a lower-dimensional space, reducing noise and facilitating model training. Additionally, using only past match data ensures a realistic setup by avoiding data leakage.

In contrast, LSTM and Transformer models were designed to exploit the sequential nature of the data. Their input consists of time series representing team performance evolution across previous matches. Preprocessing focused on sequence generation and normalization, deliberately avoiding dimensionality reduction to preserve temporal structure. A window of the previous five matches was used for these models.

All models were trained and evaluated using a consistent validation framework, employing multiclass classification metrics to compare performance.

Results and Conclusions

Results

To establish a reference point, several baseline metrics were calculated, as shown in Table 1.

Tabla 1: Baseline Accuracy

Strategy	**Accuracy**
Always predict home win	0.4464
Always predict away win	0.3006
Always predict draw	0.2530
Weighted random	0.3624

The DNN model achieved an overall accuracy of 57 % on 3,253 matches, outperforming baseline approaches. Class-level analysis shows stronger performance for home wins—F1 = 0.64—and away wins—F1 = 0.58—while draws remain more difficult—F1 = 0.45.

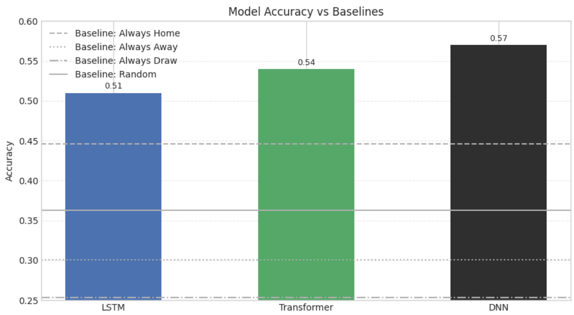
The LSTM model achieved an overall accuracy of 51 %, also outperforming the baseline but to a lesser extent. It follows the same trend, with better performance for home and away classes—F1 = 0.61 and F1 = 0.54—while draw prediction is notably weaker—F1 = 0.30.

The Transformer model reached an overall accuracy of 54 %, again surpassing the baseline. Similar to previous models, it performs better on home and away classes—F1 = 0.65 and F1 = 0.56—while struggling with draws—F1 = 0.24.

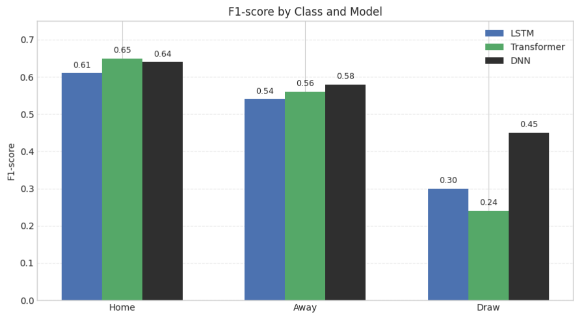
Conclusions and Future Work

After analyzing the performance of the three evaluated architectures, the deep neural network emerges as the most suitable approach. It not only achieves the highest overall accuracy but also shows a significant advantage in detecting draws, which represent the main challenge due to class imbalance. Specifically, the DNN outperforms the LSTM by 15 points and the Transformer by 21 points in predicting draws.

Figure 1 (left) shows a comparison of overall model performance against baseline approaches, while Figure 1 (right) presents a comparison of F1-scores by class.



Model accuracy vs baseline



F1-score by class and model

Figura 1: Performance comparison: accuracy (left) and F1-score (right)

The results confirm that draw prediction is the main bottleneck of the problem. Improving performance on this class is key to developing more robust and reliable models for sports analytics. This could help identify factors influencing match outcomes and enable more accurate predictions. Additionally, this work opens the possibility of exploring betting applications by combining predictive models with economic strategies to potentially outperform market efficiency.